

Body-fitted finite-volume NACA0012 verification

Actual body-fitted Gmsh meshes; strict physical relative error <3%.

Scheme: {'momentum_convection': 'linear-upwind', 'pressure_gradient': 'gauss-linear', 'face_interpolation': 'projected-linear', 'wall_gradie

CPU calculations; no new GPU or matched TensorLBM result.

Positive mesh geometry and nonorthogonality gate precede all solves.

scale-3: 2190 cells; converged=True; accepted=False

{'drag': 0.12928968858768622, 'lift': 0.21346753343055289}

{'drag_relative_error': 0.0350914, 'lift_relative_error': 0.0627487}

scale-2: 4832 cells; converged=True; accepted=False

{'drag': 0.1278956900099021, 'lift': 0.21557237337687038}

{'drag_relative_error': 0.023931, 'lift_relative_error': 0.0732277}

scale-1: 18712 cells; converged=True; accepted=True

{'drag': 0.12210515141245215, 'lift': 0.19605037112024748}

{'drag_relative_error': 0.0239907, 'lift_relative_error': 0.0249336}

Full independently replayed equations: True

Refinement sequence accepted: False

Asymptotic grid regime qualified: False

Problem, numerical method and reproducibility

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Problem and reference

NACA0012 at $Re=1000$, incidence 4 degrees, unit chord and the reference 36C by 16C domain. The quarter-chord point is (12C,8C). The body is rotated clockwise by 4 degrees about that point, while inlet velocity is (1,0) m/s; drag and lift are global x/y forces in these physical freestream axes. The inlet has uniform velocity; the outlet and open top/bottom boundaries have zero pressure, zero-gradient outflow velocity, and freestream velocity on backflow. The paper specifies open boundaries but does not give an exact FV formula, so this pressure-open implementation is an explicit modeling choice. Geometry/domain values are independently read from [Di Ilio et al. figure 3](<https://arxiv.org/html/2006.10487>). Figures 10 and 11 give digitized coefficient references and ± 0.0001 graph-reading intervals; worst endpoint errors are used. The domain and boundary review is archived in docs/naca-reference-domain-review.json.

Mesh and discretization

Gmsh 4.15.2 frontal Delaunay generates conforming triangles with physical boundary tags; cylinder CAD arcs or a closed NACA0012 interpolating spline define the body. Physical wall faces are linear chords. See the [Gmsh official manual](<https://gmsh.info/doc/texinfo/gmsh.html>) for algorithm 6 and Distance/Threshold fields. Neither Cartesian obstacle masks nor staircase boundaries are used. Positive volumes and projected face distances and a maximum nonorthogonality below 70 degrees are required before solving. The 70-degree project gate is a screening criterion, not proof of sufficient accuracy.

The recorded discretization is ``{"boundary": "pressure-open-freestream-backflow", "face_interpolation": "projected-linear", "momentum_convection": "linear-upwind", "pressure_gradient": "gauss-linear", "wall_gradient": "displacement-correction"}``. Linear upwind uses the least-squares gradient of the upstream cell and the actual face-to-cell displacement; its conservative deferred correction is inspired by [OpenFOAM linearUpwind](<https://github.com/OpenFOAM/OpenFOAM-dev/blob/master/src/finiteVolume/interpolation/surfaceInterpolation/schemes/linearUpwind/linearUpwind.C>). This is unlimited reconstruction, so no bounded/TVD claim is made. Gauss pressure gradients use geometric face interpolation and the same boundary face pressure used by pressure traction. Nonorthogonal diffusion, physical Rhie-Chow flux and no-slip reconstructed molecular stress are retained. Cylinder acceleration uses a full-grid coupled Picard matrix with an independent response check; NACA uses SIMPLE and Anderson mixing. CPU sparse solves check true residuals. No reference value enters the equations, initialization or relaxation.

Reproduction and acceptance

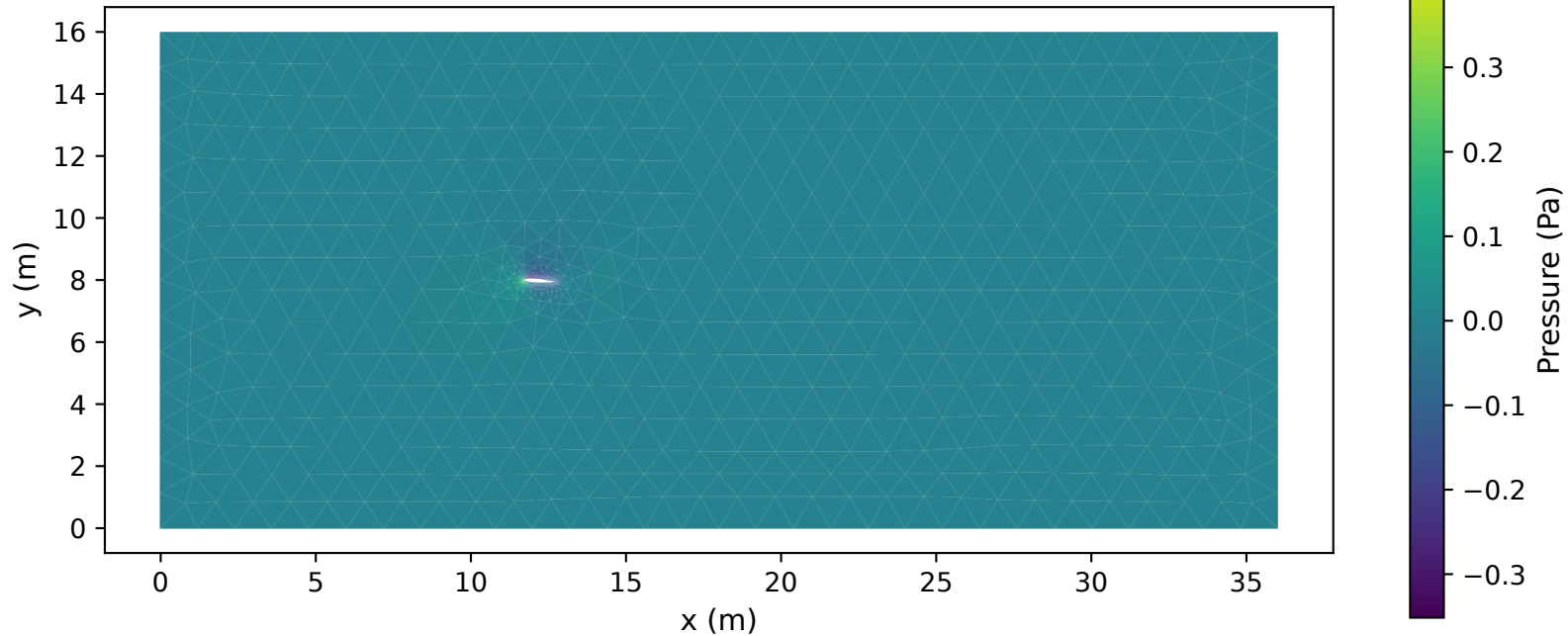
```
```bash
uv pip install --python /path/to/python gmsh numpy scipy matplotlib
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 PYTHONPATH=src python -m
tensorfvm.verification.external_matched_naca --scales 3.0 2.0 1.0 --max-iterations 2500 --output
results/naca
PYTHONPATH=src python scripts/audit_external_equations.py results/naca
PYTHONPATH=src python -m tensorfvm.verification.external_mesh_report results/naca
```
```

Raw mesh/field files, velocity/pressure arrays, face mass fluxes, separate tractions, full histories, linear residuals and SHA256 source/artifact manifests accompany this report. Each physical relative error must be strictly below 3%; steady equation residuals must be below $1e-6$ and the physical flux fixed-point defect below $1e-7$. Source-preserving independent NumPy replay checks full configured equations. Three runs do not establish an asymptotic grid regime; qualification of the finest run does not qualify the entire refinement sequence. GPU and matched TensorLBM runs have not been performed for this new unstructured path.

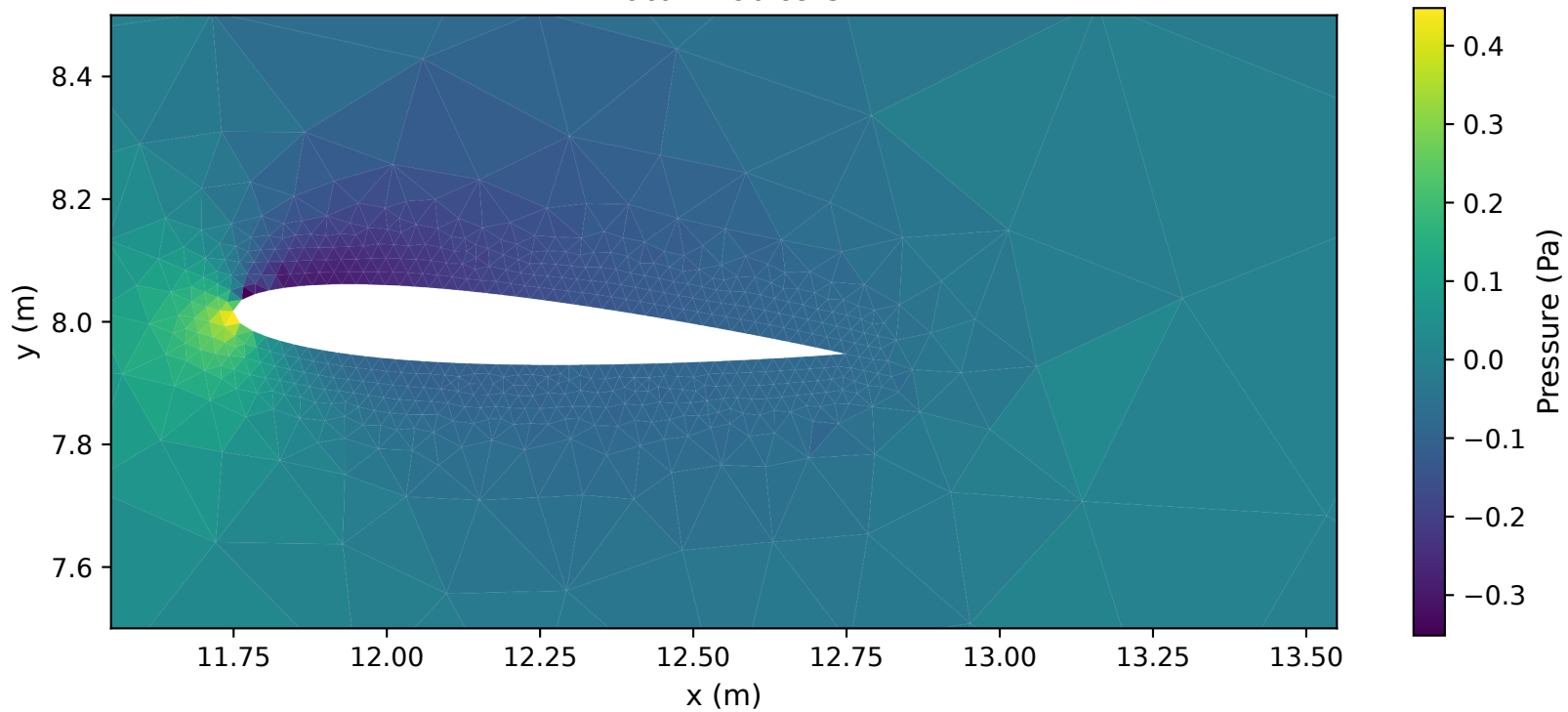
The finest NACA run reached the initial 2500-iteration cap above $1e-6$, so it was not accepted. Continue its actual saved fields using `PYTHONPATH=src python scripts/continue_matched_naca.py --output results/naca --extra-iterations 1500``. This preserves `results/naca-initial`` and appends the true iteration history. The published continuation needed 135 additional steps, for 2635 total, and then passed the same unchanged steady and physical gates. The initial stage, restart field SHA256 and both producer snapshots are retained. The command block above reproduces the initial stage; this continuation step reproduces the final run.

Problem, numerical method and reproducibility

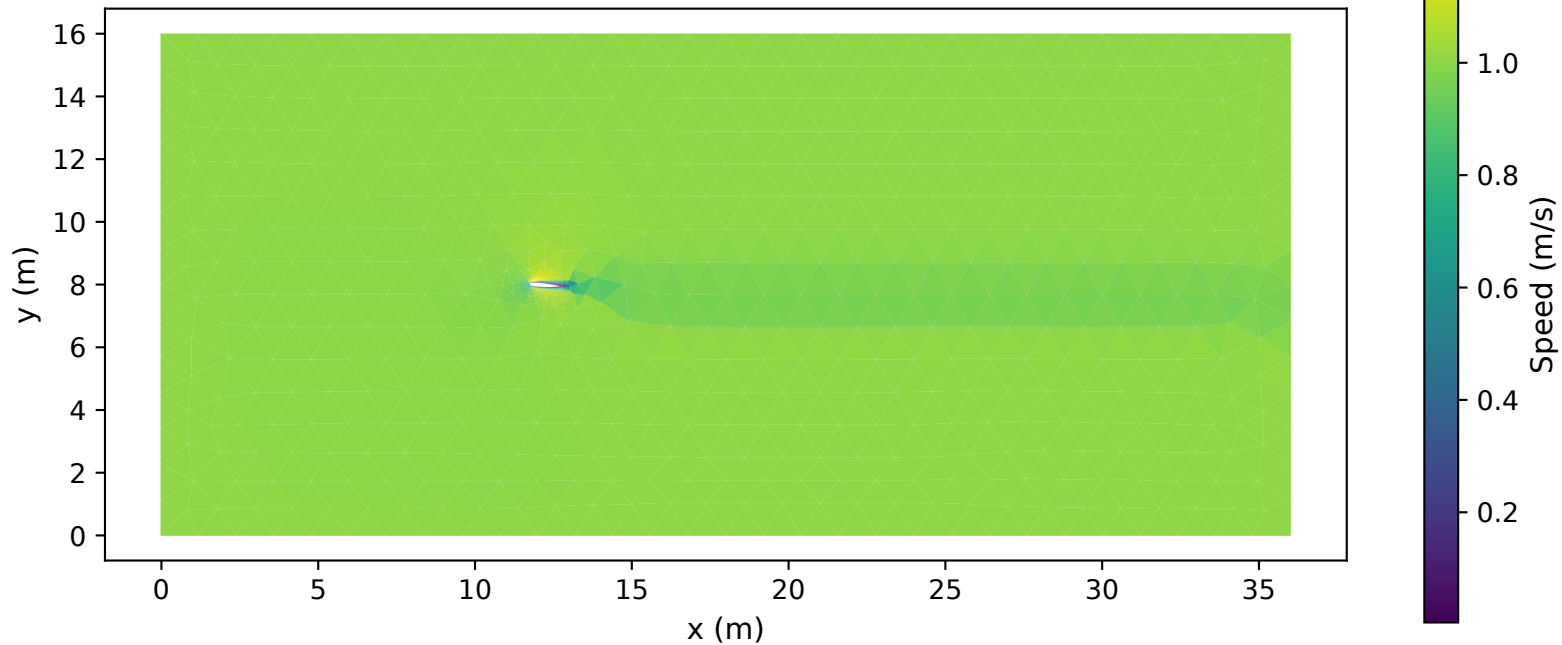
naca: 2190 cells



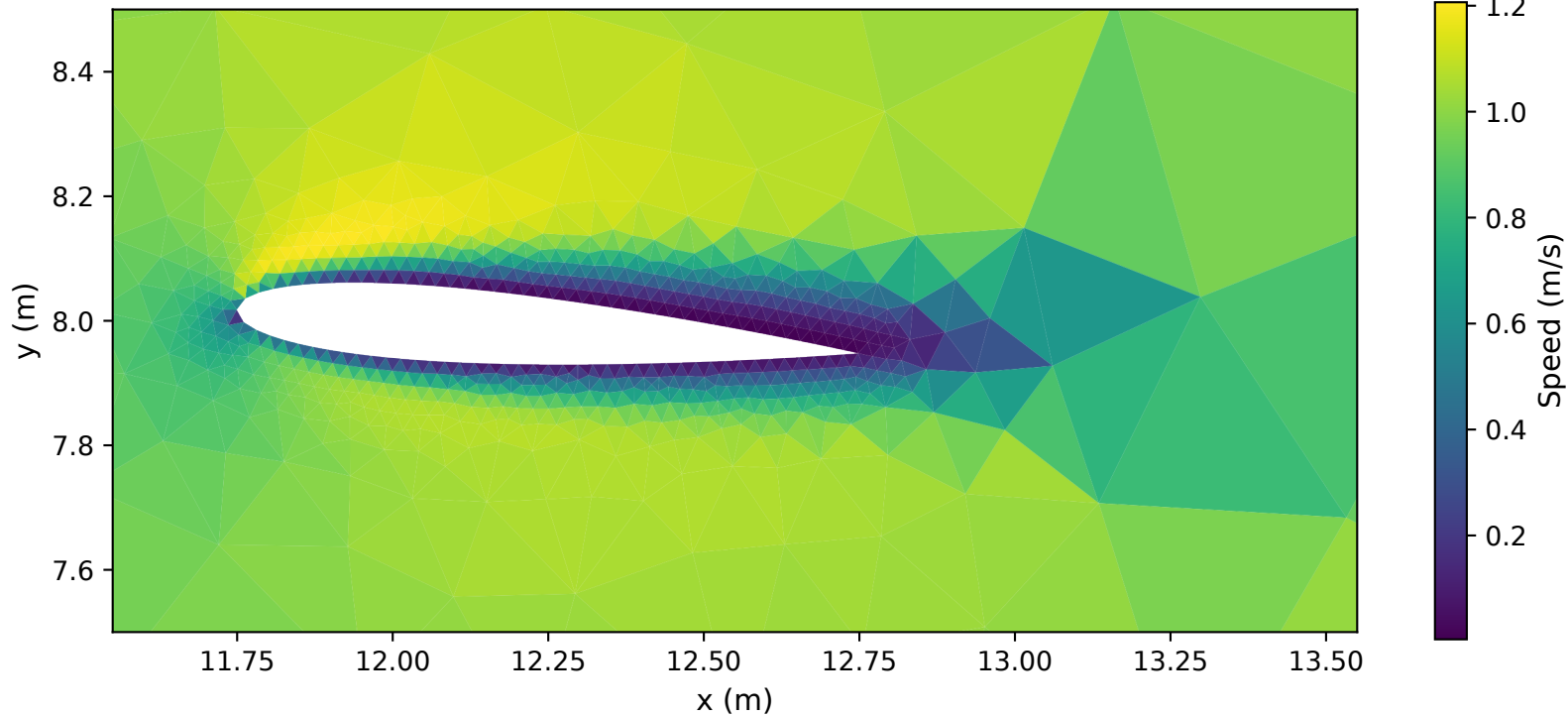
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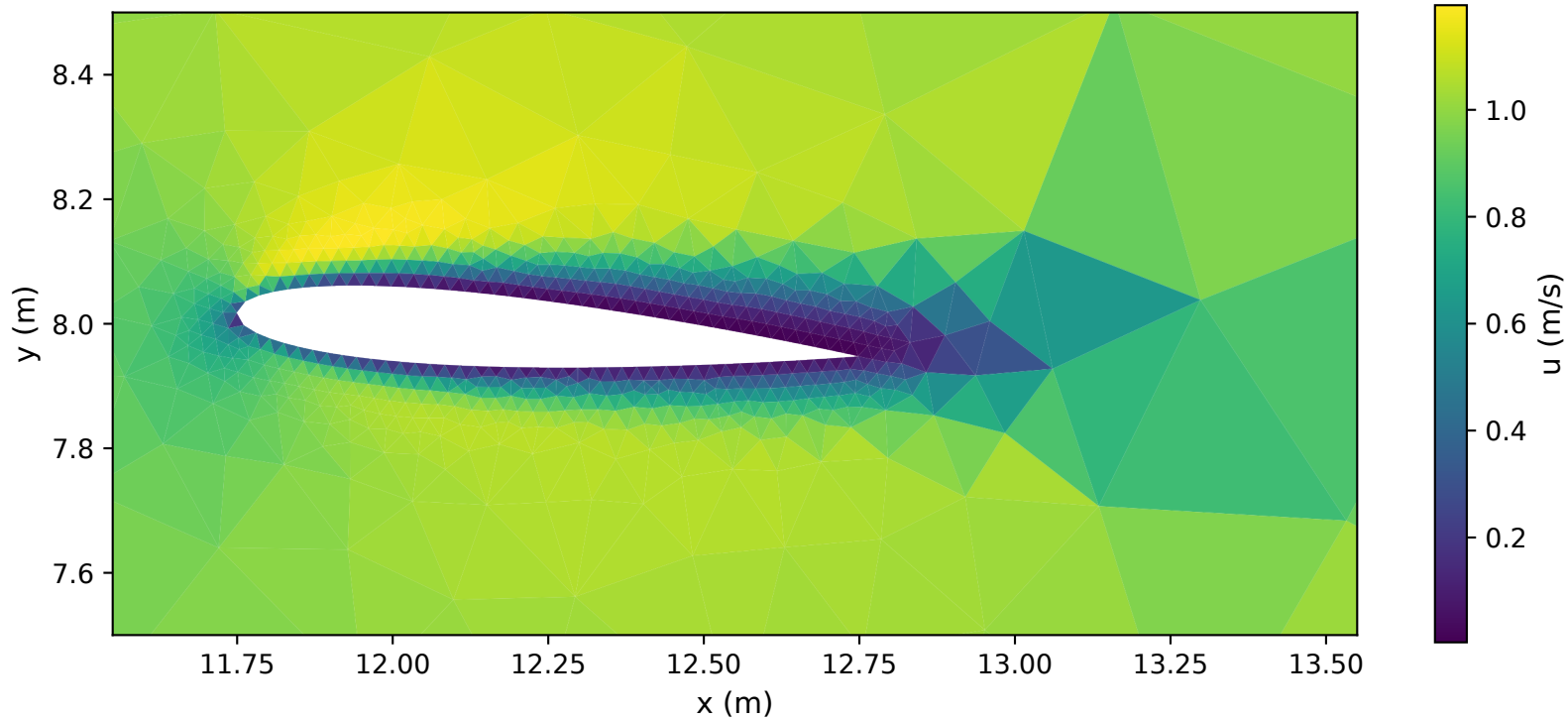
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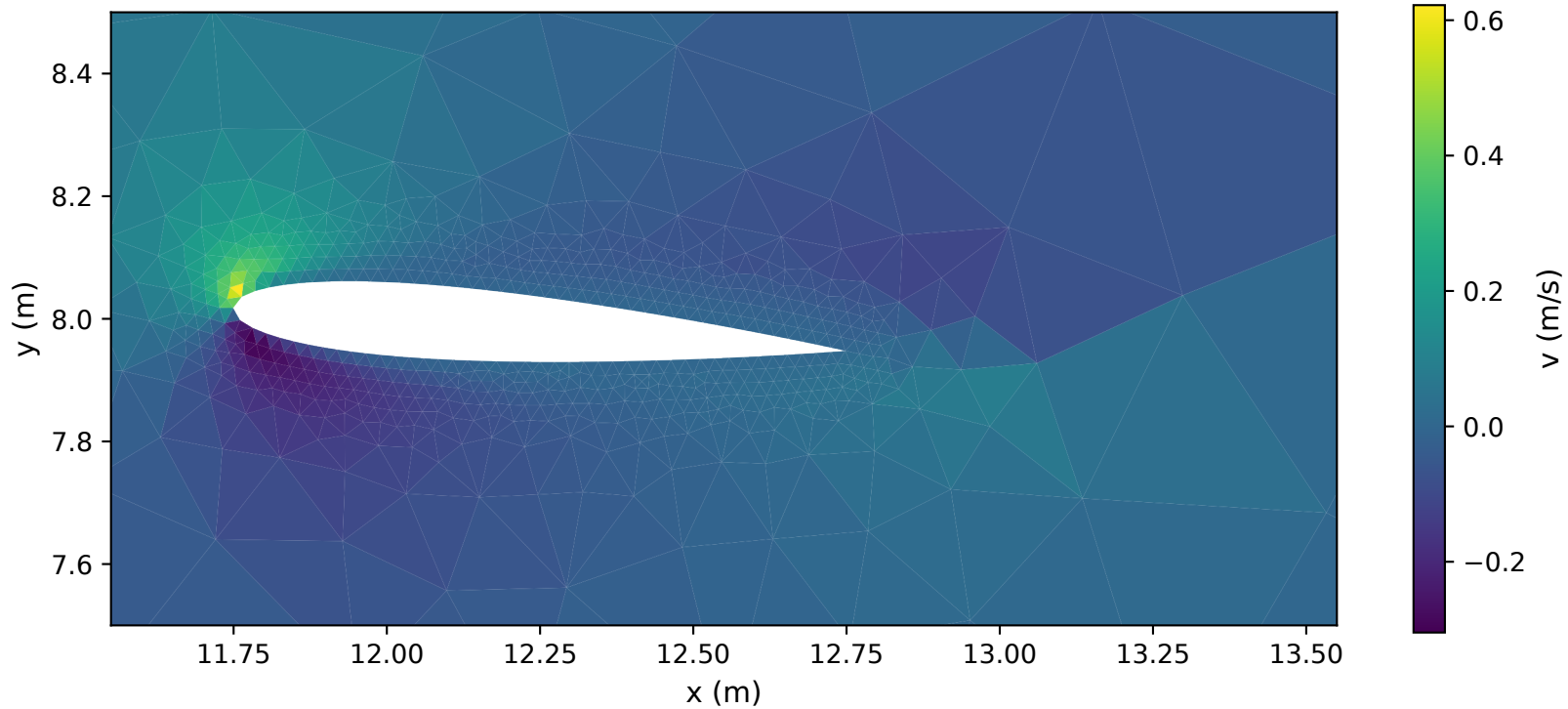
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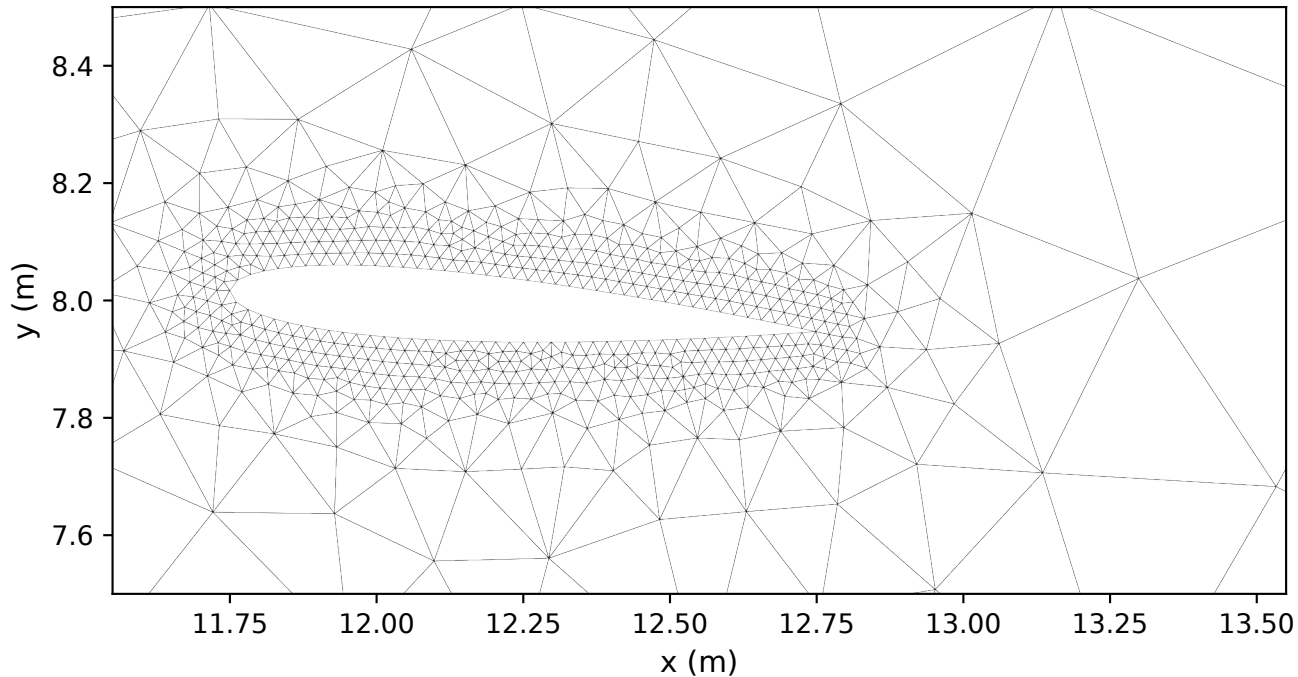
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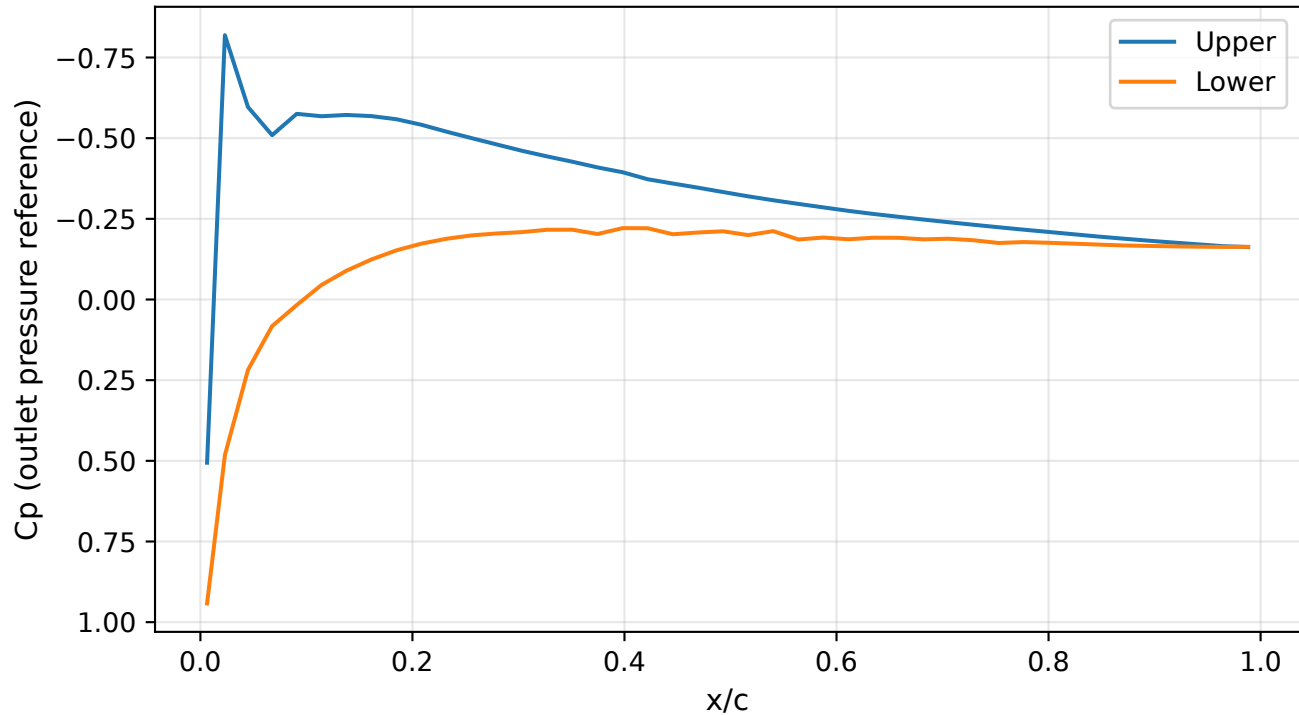


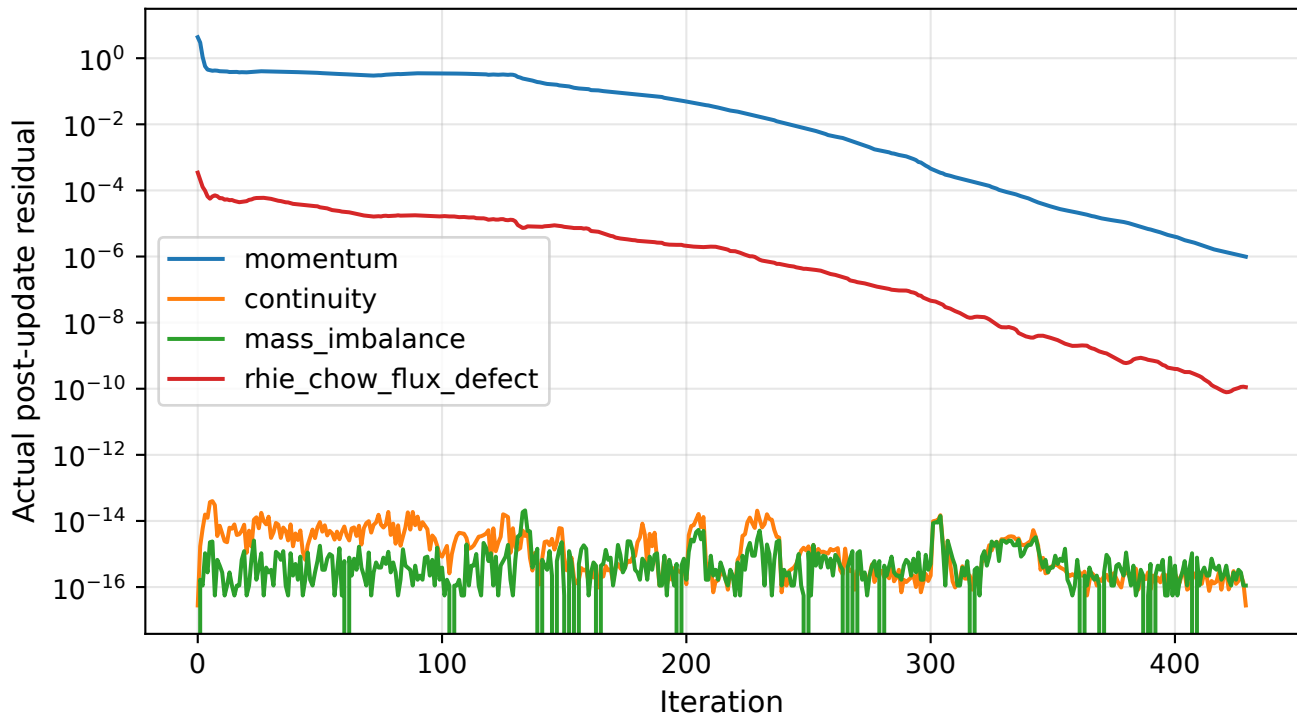
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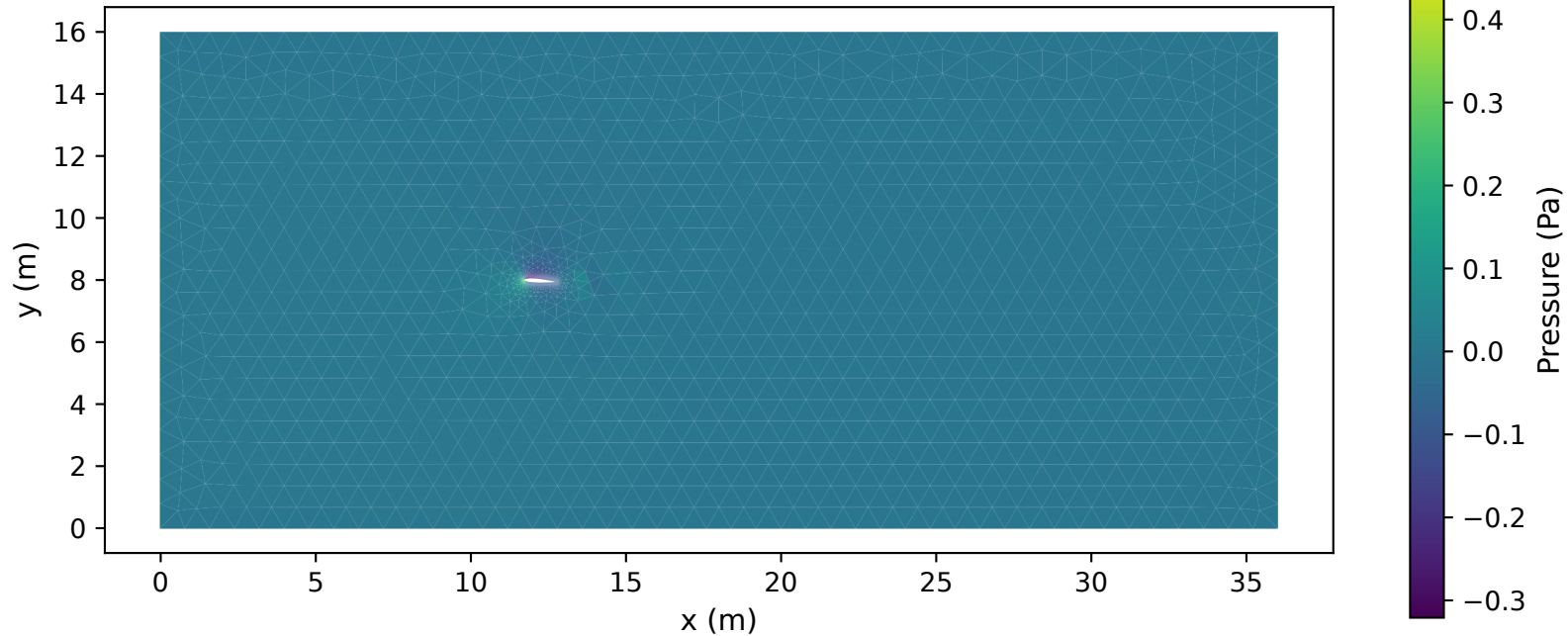
Actual body-fitted mesh



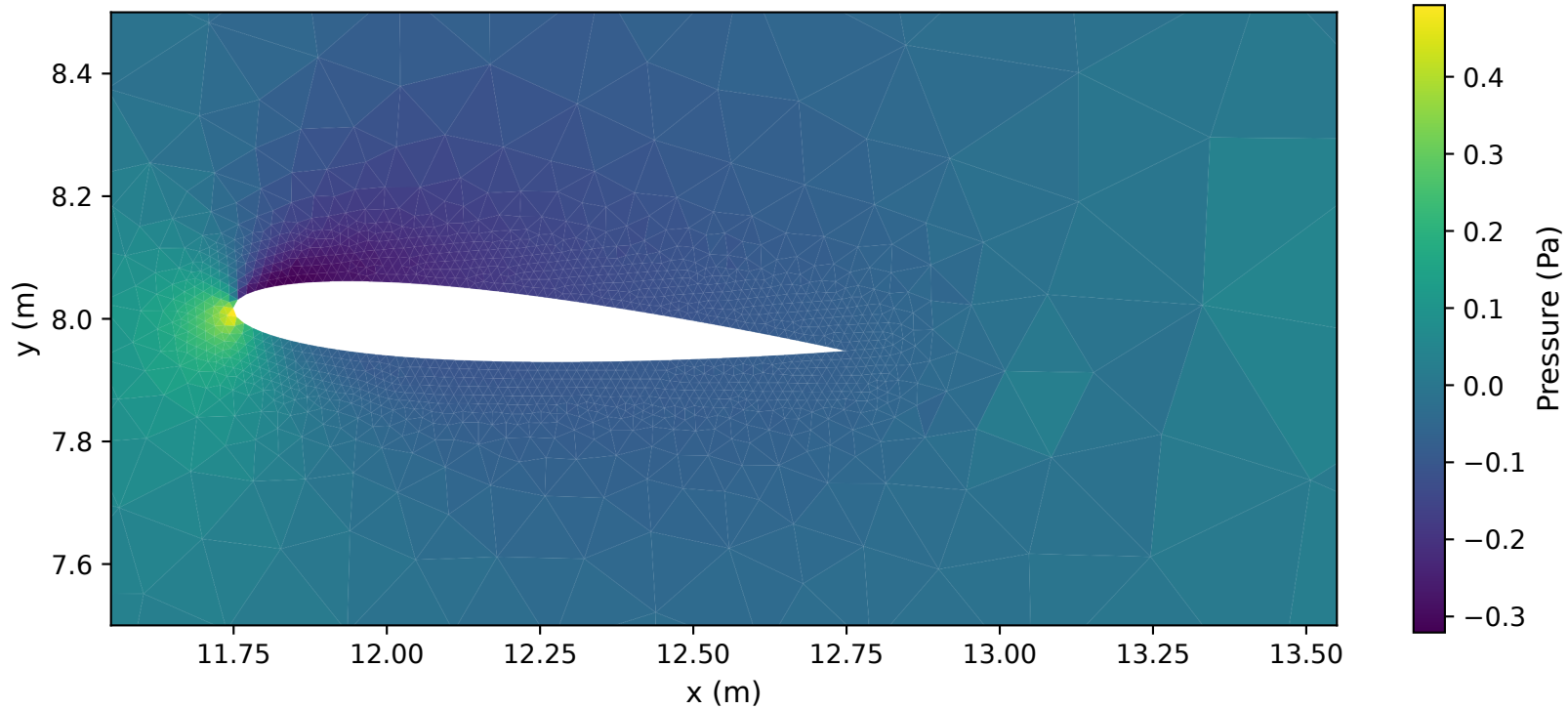




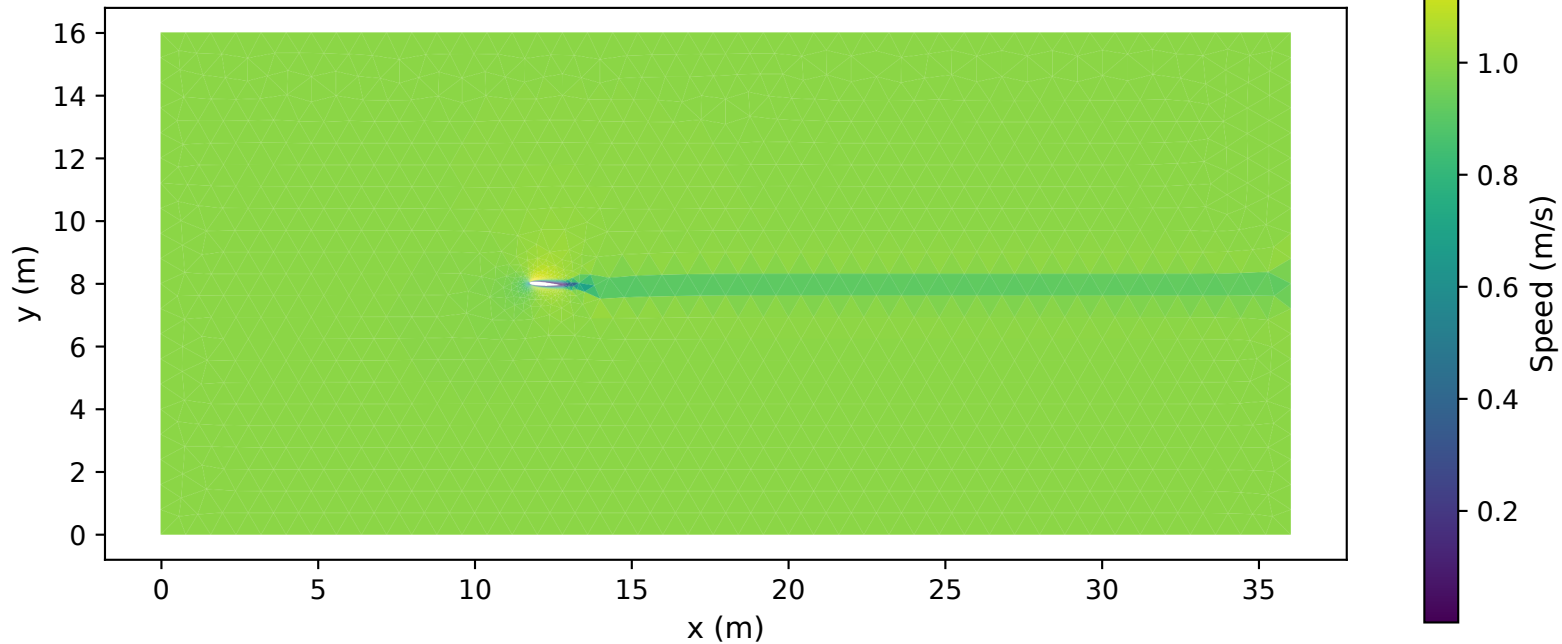
naca: 4832 cells



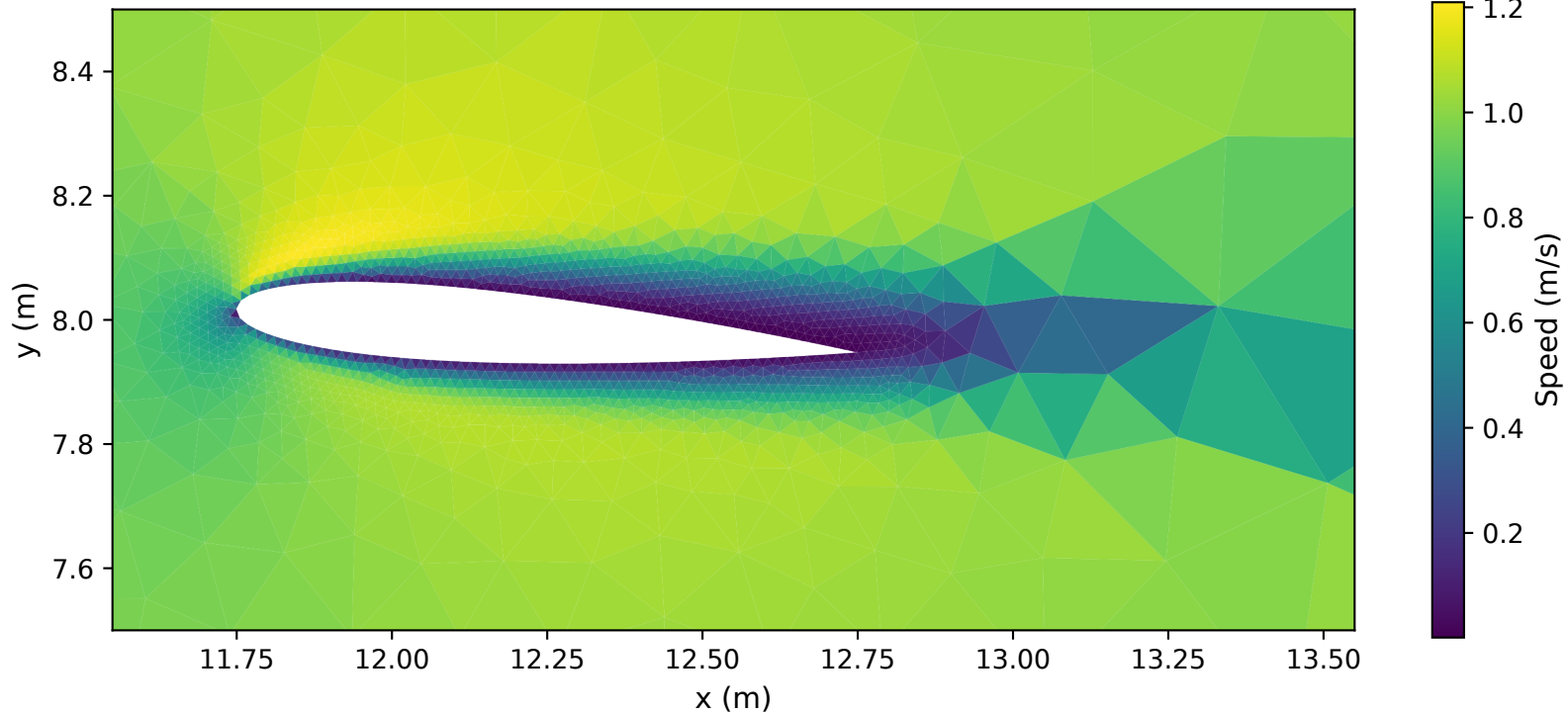
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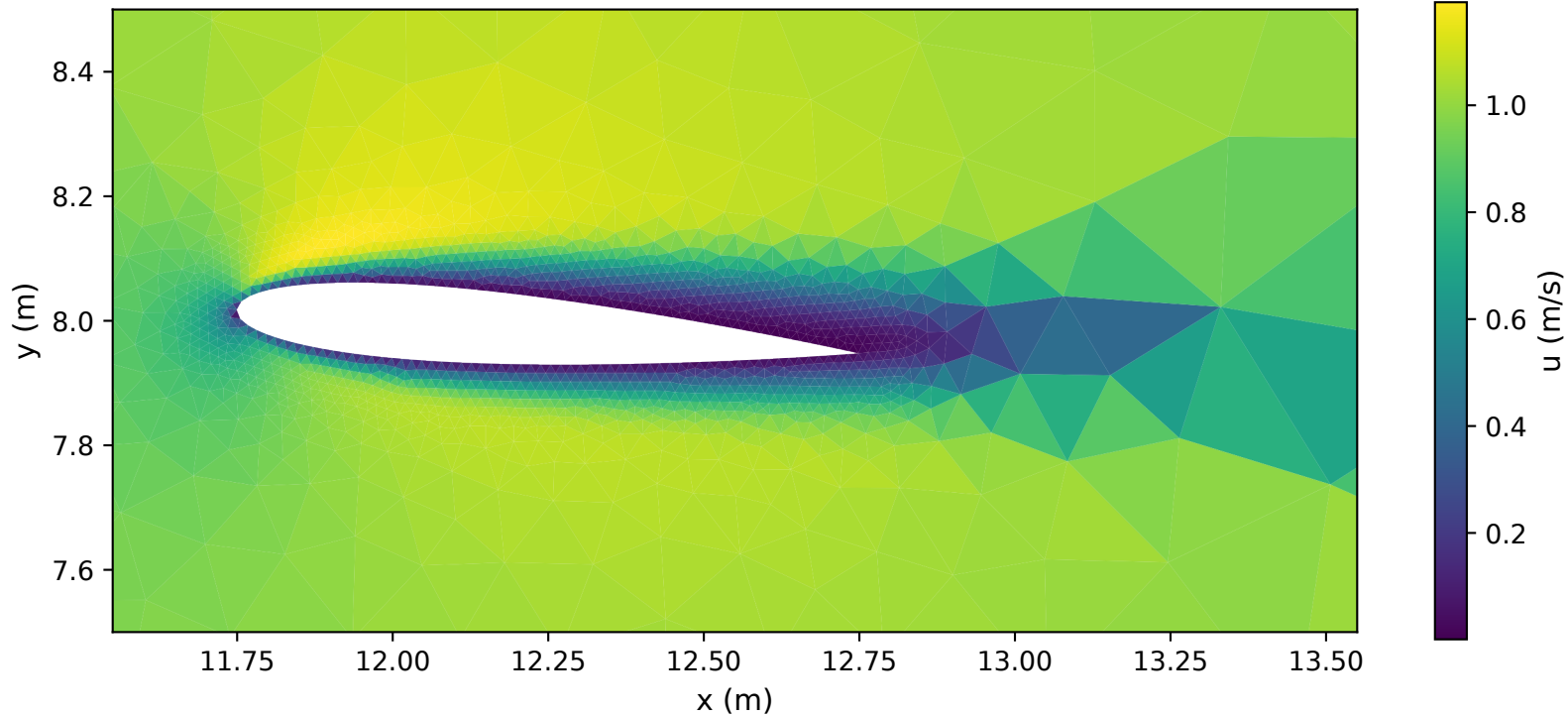
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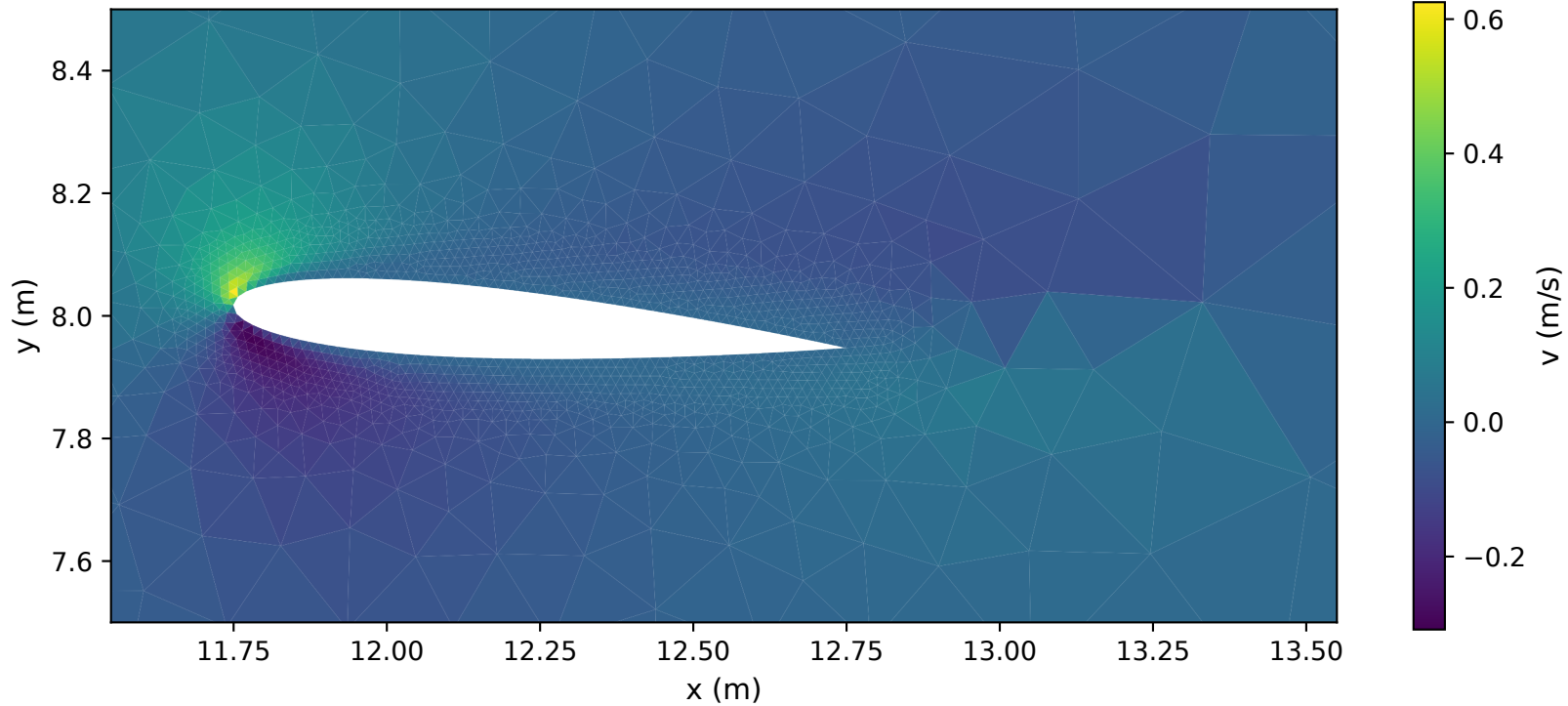
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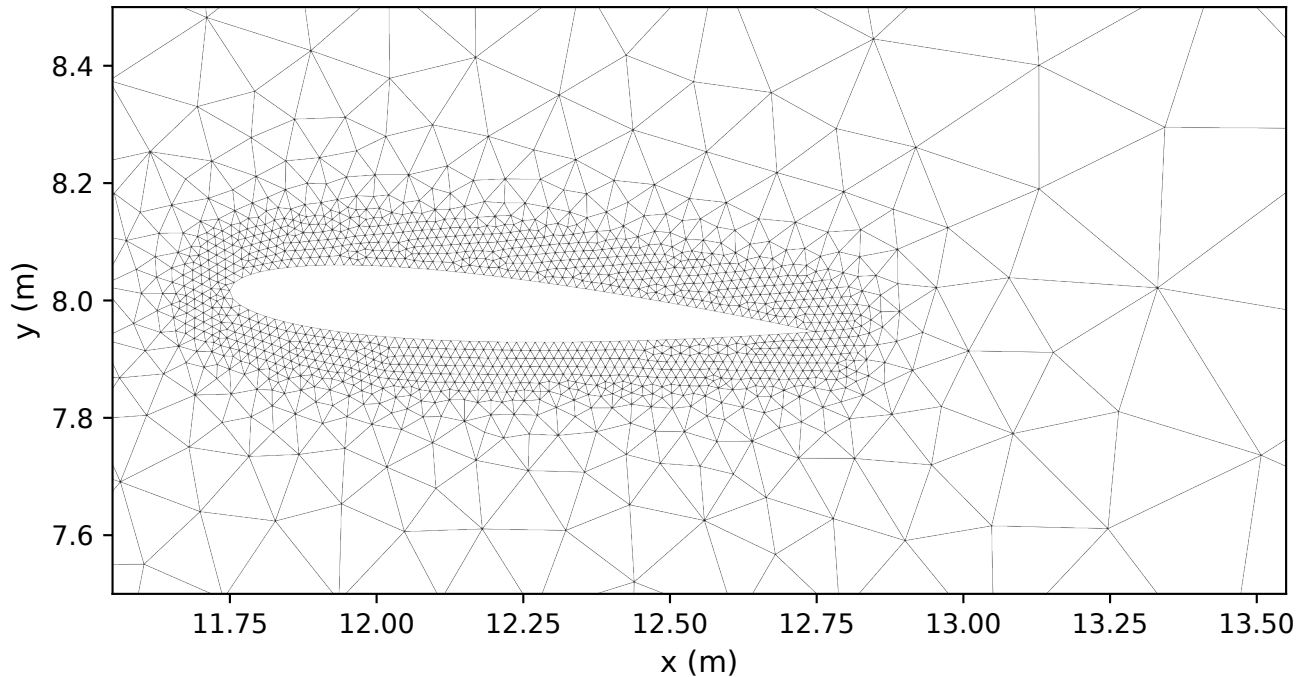
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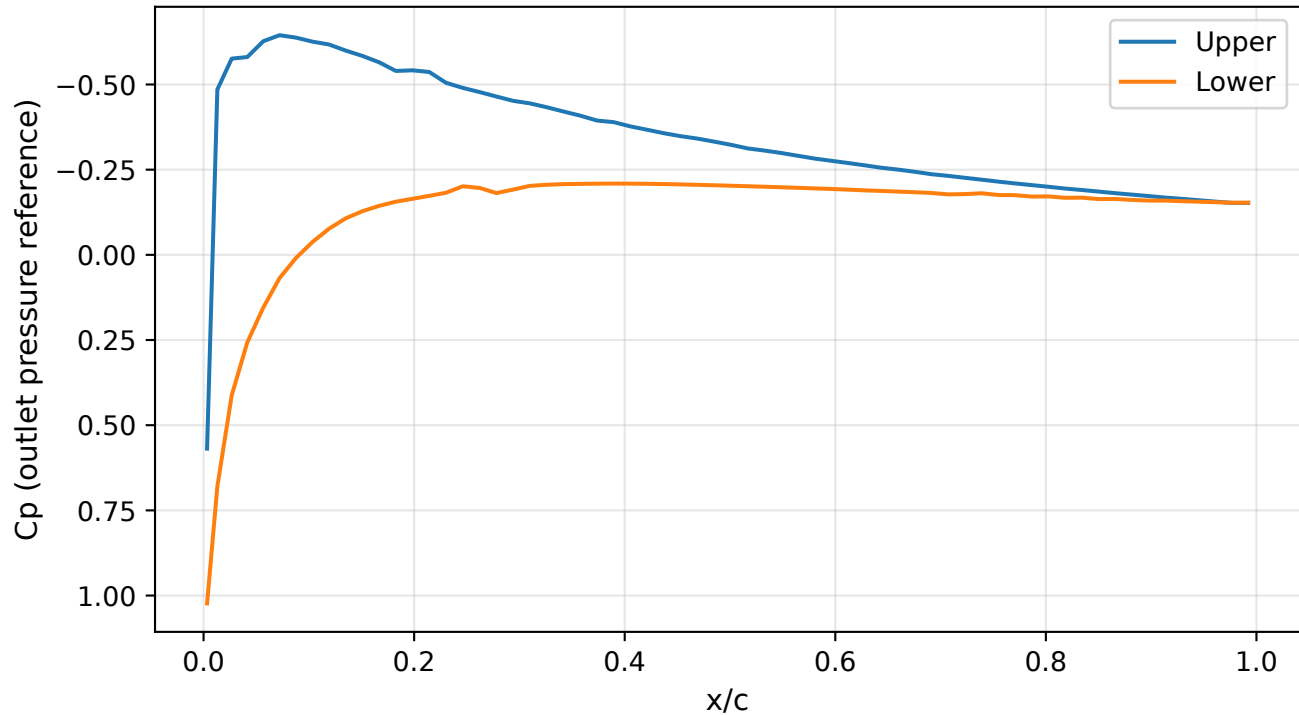


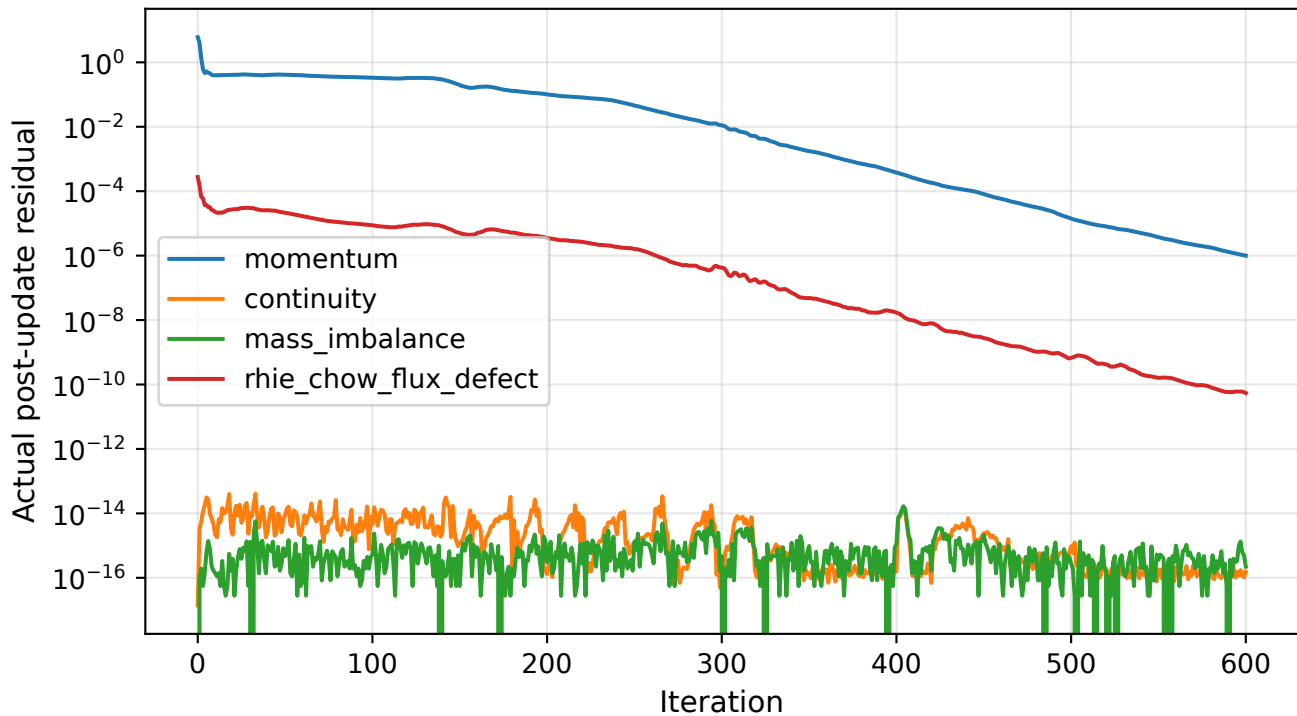
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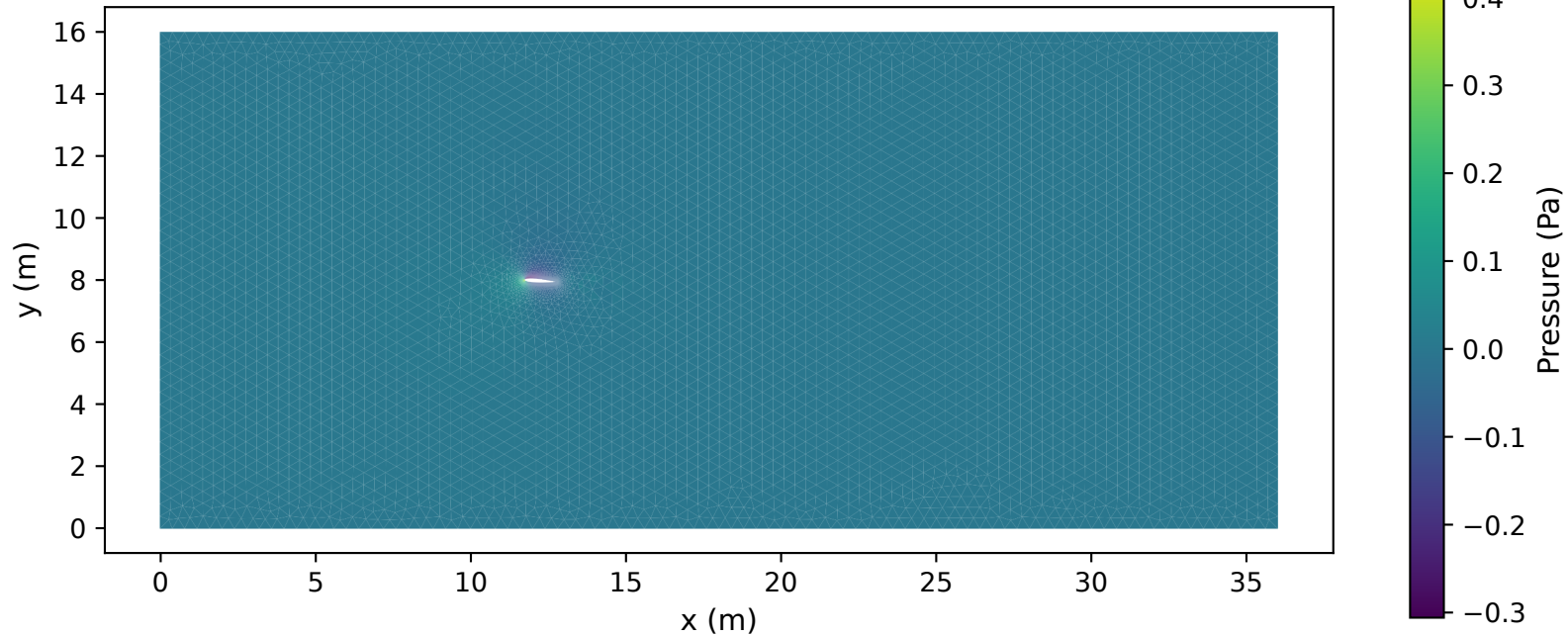
Actual body-fitted mesh



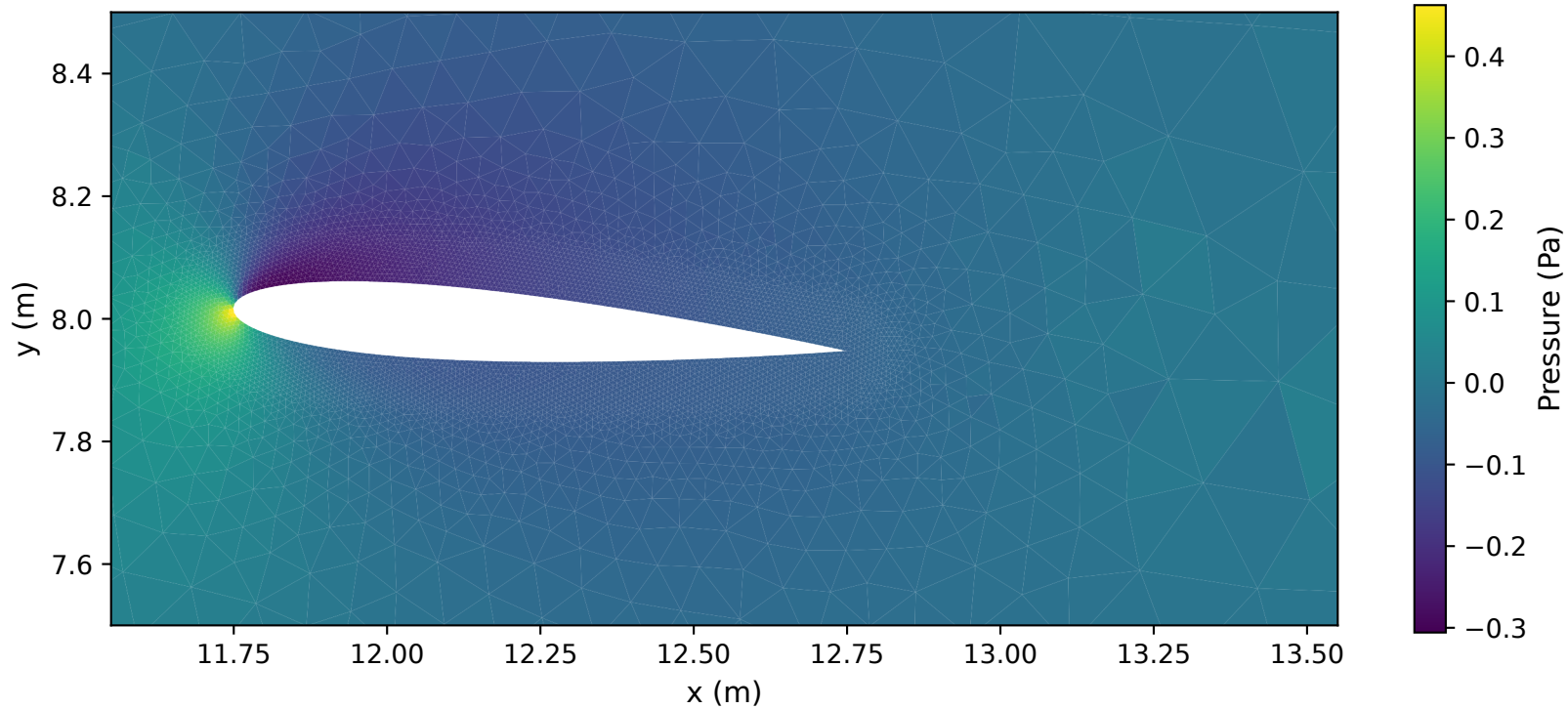




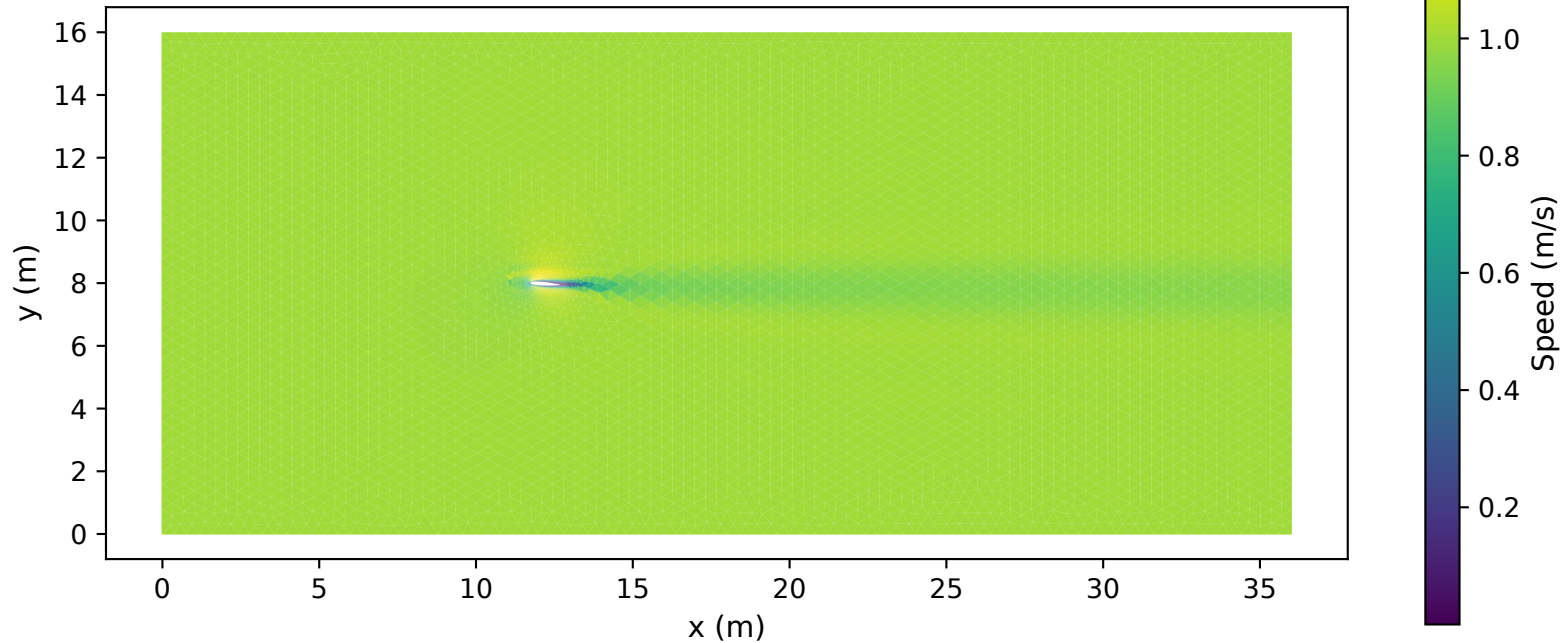
naca: 18712 cells



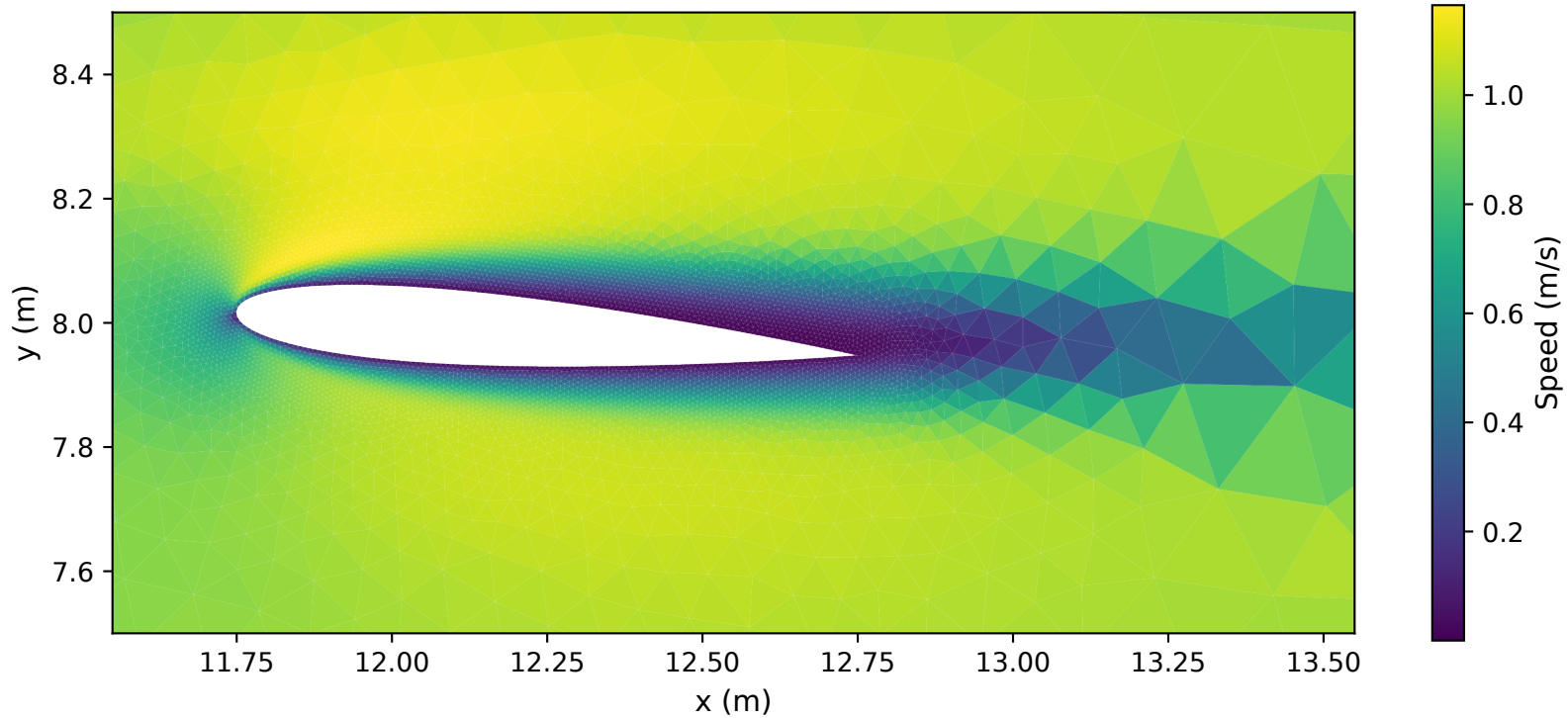
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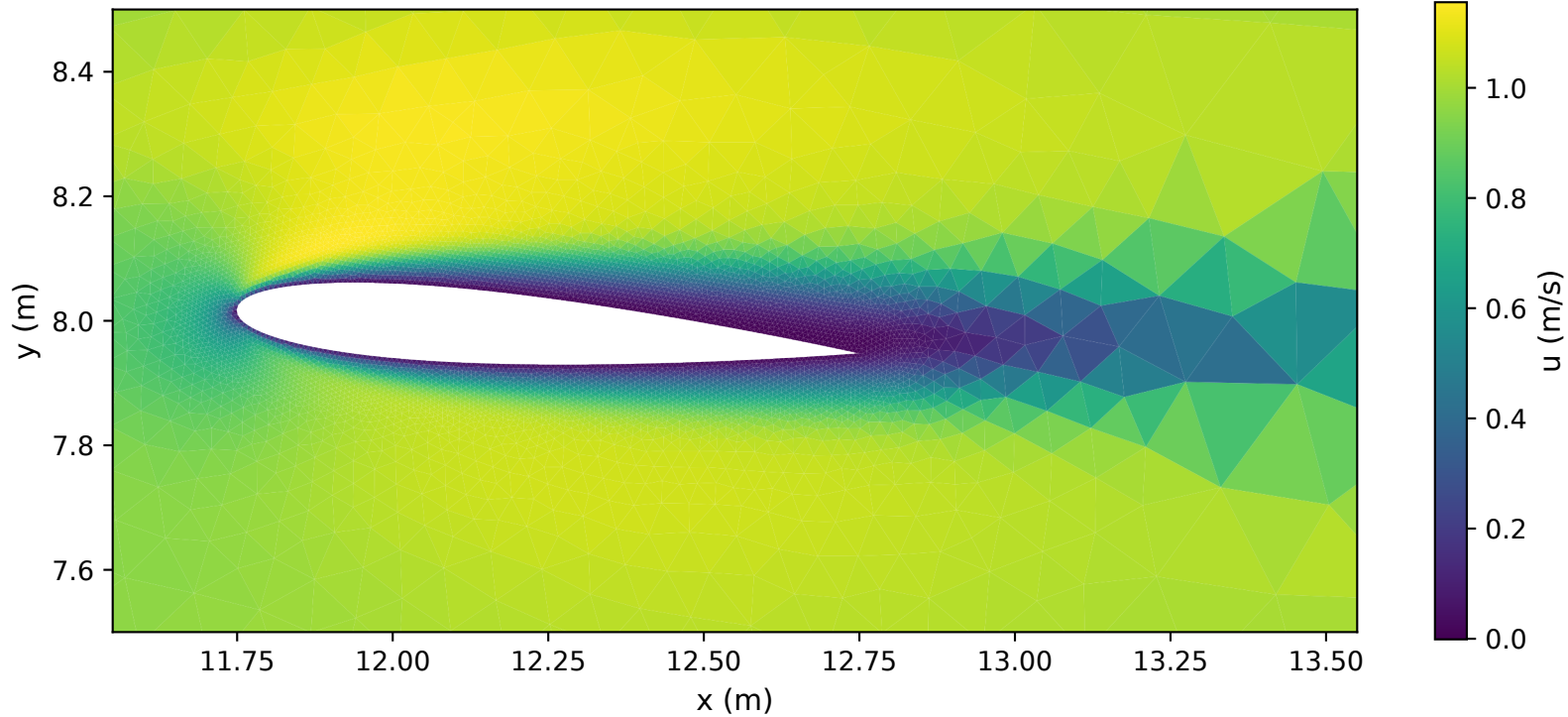
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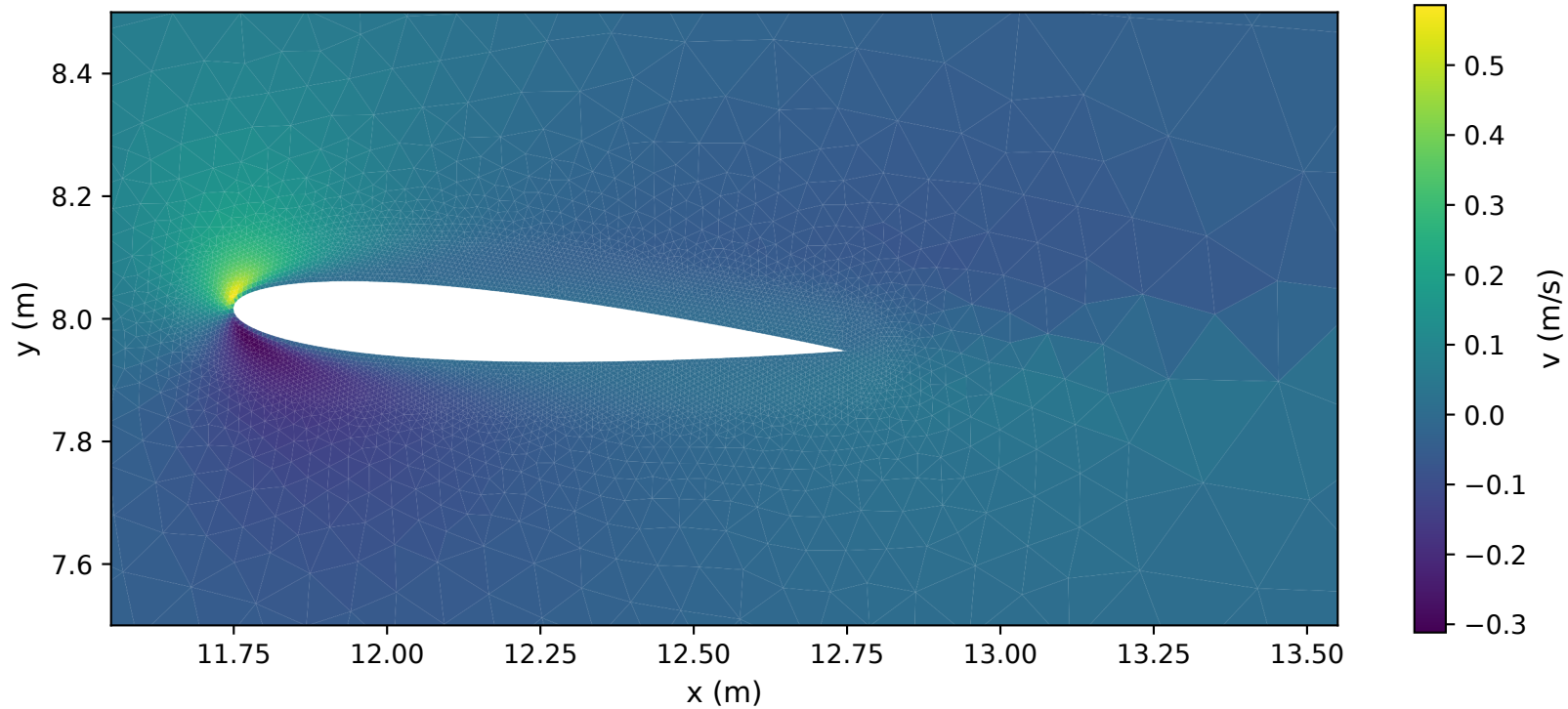
naca: 18712 cells



naca: 18712 cells



naca: 18712 cells



Actual body-fitted mesh

